

Constraint Analysis of Electromagnetic Stress Asymmetry in Resonant Cavities: Upper Bounds from Emergent Vacuum Response Theory

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Abstract

This work presents a constraint analysis of electromagnetic stress asymmetry in asymmetric resonant cavity systems. Building on Emergent Vacuum Response Theory (EVRT), which provides a phenomenological framework for nonequilibrium electromagnetic stress redistribution, we derive quantitative upper bounds on any effective stress-asymmetry response using experimentally relevant sensitivity limits.

Rather than asserting anomalous-force phenomena, the analysis follows standard physical methodology by bounding an effective response parameter through noise-floor considerations, scaling relations, and representative resonant cavity parameters. For typical tabletop configurations, the effective response parameter is constrained to $\chi_{\text{eff}} \lesssim 10^{-17}$, indicating that any electromagnetic stress asymmetry effect must be extremely small under current experimental conditions.

This work represents the constraint stage of the EVRT research program, linking prior theoretical modeling and numerical simulations to experimentally testable limits. By converting EVRT into a quantitatively constrained hypothesis, the results define the accessible parameter space for resonant cavity experiments and establish a clear pathway for systematic falsification or verification.

1 Introduction

Emergent Vacuum Response Theory (EVRT) has been proposed as a phenomenological framework for investigating whether structured, nonequilibrium electromagnetic systems may exhibit measurable stress-energy redistribution under asymmetric boundary conditions. Previous work has established scaling relations, simulation-based predictions, and experimental protocols for testing such effects.

However, in the absence of definitive experimental confirmation, it is essential to quantify the limits within which any such effect could exist. In physics, null results and measurement sensitivity play a central role in constraining theoretical models. The present work adopts this approach by establishing upper bounds on the effective response parameters associated with EVRT-like scaling behavior.

This analysis does not assume the existence of a new physical interaction. Instead, it treats EVRT as a phenomenological model and determines the range of parameter values consistent with experimentally relevant sensitivity limits.

This work builds on prior development of EVRT scaling relations, numerical simulations, and experimental protocols, and focuses specifically on constraining the allowable parameter space using experimentally relevant sensitivity limits.

2 Phenomenological Framework

Following prior work, the electromagnetic energy density is expressed as

$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0} B^2. \quad (1)$$

A dimensionless asymmetry parameter is defined as

$$A = \frac{\int (u_+ - u_-) dV}{\int (u_+ + u_-) dV}. \quad (2)$$

A representative phenomenological scaling relation for force is written as

$$F \sim \chi_{\text{eff}} A (E^2 + B^2), \quad (3)$$

where χ_{eff} is an effective response parameter incorporating geometry, boundary conditions, and coupling effects. This expression is not derived from first principles but serves as a scaling relation for generating testable predictions.

For resonant systems, the characteristic field-energy scale may be represented phenomenologically by the cavity energy enhancement,

$$E^2 + B^2 \propto \frac{PQ}{V}, \quad (4)$$

where P is input power, Q is the resonator quality factor, and V is the effective cavity volume. This gives the reduced constraint model

$$F \sim \chi_{\text{eff}} A \frac{PQ}{V}. \quad (5)$$

3 Constraint Methodology

To establish upper bounds, we consider the requirement that any unconfirmed force signal must lie below the experimental noise floor:

$$F_{\text{obs}} < F_{\text{noise}}. \quad (6)$$

Substituting the reduced phenomenological scaling relation yields

$$\chi_{\text{eff}} < \frac{F_{\text{noise}} V}{APQ}. \quad (7)$$

This inequality provides a direct upper bound on the effective response parameter given experimental sensitivity and system parameters. The value of χ_{eff} should therefore be interpreted as a constrained effective parameter, not as a universal constant.

4 Representative Parameter Space

We consider representative values for tabletop resonant cavity systems:

- Input power: $P \sim 10\text{--}100$ W
- Quality factor: $Q \sim 10^4$
- Effective volume: $V \sim 1.5 \times 10^{-4}$ m³
- Asymmetry parameter: $A \sim 0.05\text{--}0.15$
- Noise floor: $F_{\text{noise}} \sim 10\text{--}100$ nN

For representative system parameters $P = 50$ W, $Q = 10^4$, $V = 1.5 \times 10^{-4}$ m³, $A = 0.1$, and $F_{\text{noise}} = 20$ nN, the effective response parameter is bounded by

$$\chi_{\text{eff}} \lesssim 6 \times 10^{-17}. \quad (8)$$

This value is illustrative and depends on system configuration, but it demonstrates that any EVRT-like effect must be extremely small under current sensitivity assumptions.

5 Results

Figure 1 shows how the upper bound on χ_{eff} depends on the experimental noise floor. The relationship is linear: improving force sensitivity by one order of magnitude tightens the constraint by one order of magnitude.

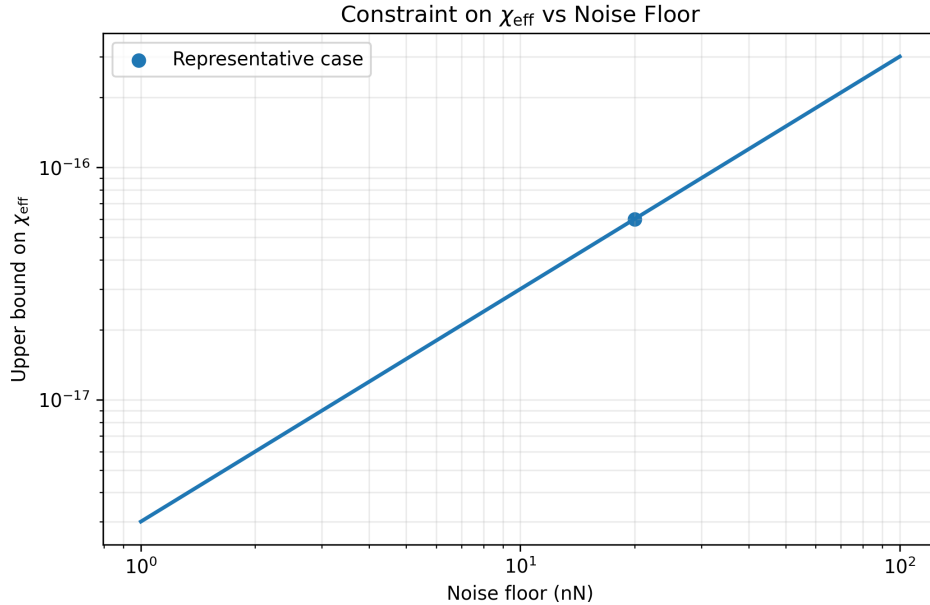


Figure 1: Upper bound on χ_{eff} as a function of experimental noise floor for fixed system parameters $P = 50$ W, $Q = 10^4$, $V = 1.5 \times 10^{-4}$ m³, and $A = 0.1$. Improved sensitivity directly tightens constraints.

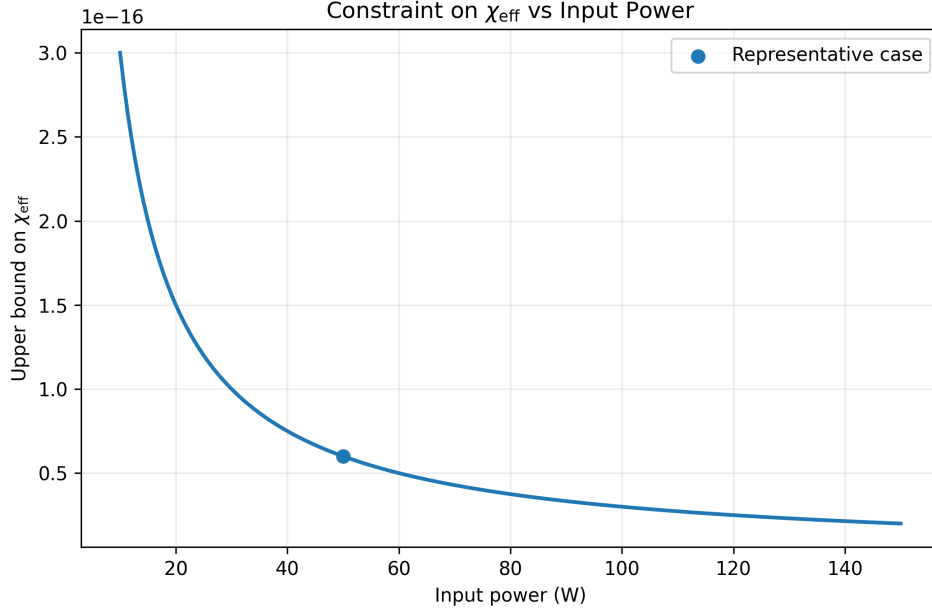


Figure 2: Upper bound on χ_{eff} as a function of input power for fixed noise floor $F_{\text{noise}} = 20$ nN, $Q = 10^4$, $V = 1.5 \times 10^{-4} \text{ m}^3$, and $A = 0.1$. Increasing power strengthens the constraint, although practical limits are set by heating, electromagnetic coupling, and mechanical stability.

Figure 2 shows how increasing input power tightens the bound. Higher input power strengthens sensitivity to EVRT-like behavior, provided systematic effects remain controlled.

Figure 3 shows the combined dependence on asymmetry parameter and input power. The strongest bounds occur for high-power, high-asymmetry configurations, assuming the noise floor remains fixed.

Noise Floor (nN)	Power (W)	Asymmetry A	χ_{eff} Bound
100	50	0.10	3.0×10^{-16}
20	50	0.10	6.0×10^{-17}
10	100	0.10	3.0×10^{-18}
1	100	0.10	3.0×10^{-19}
1	100	0.15	2.0×10^{-19}

Table 1: Representative upper bounds on χ_{eff} under varying experimental sensitivity and drive conditions, assuming $Q = 10^4$ and $V = 1.5 \times 10^{-4} \text{ m}^3$.

6 Sensitivity Requirements

The constraint relation shows that three experimental improvements most directly tighten the allowed parameter space:

- reducing force noise floor;
- increasing controlled input power without introducing thermal or electromagnetic artifacts;

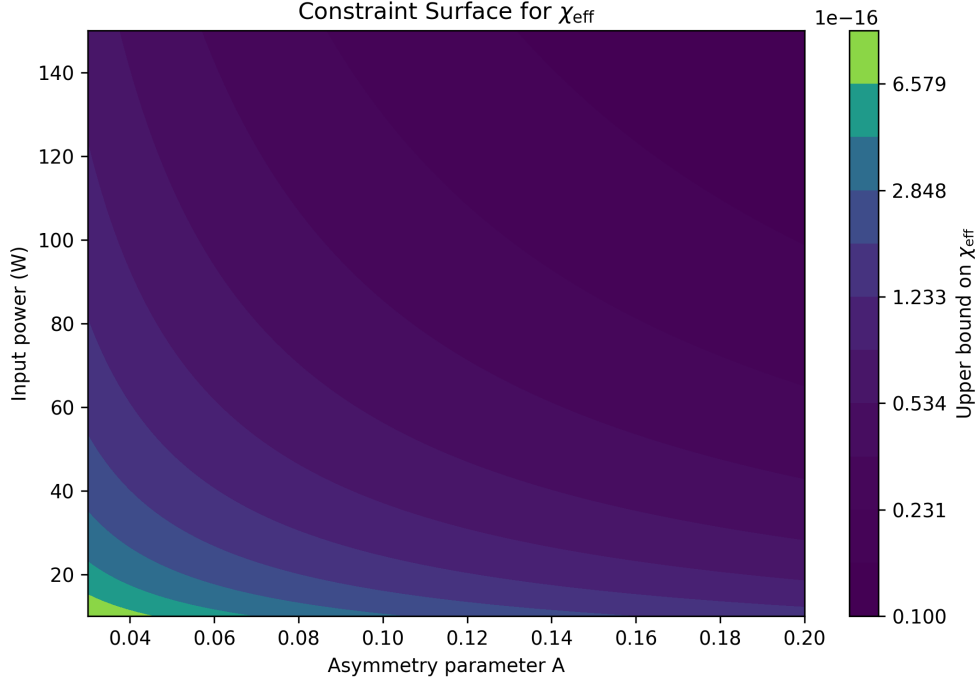


Figure 3: Constraint surface for χ_{eff} as a function of asymmetry parameter A and input power P , with $F_{\text{noise}} = 20$ nN, $Q = 10^4$, and $V = 1.5 \times 10^{-4}$ m³. Higher asymmetry and higher power produce tighter bounds.

- increasing geometric asymmetry while preserving well-characterized mode structure.

Reducing the noise floor from 20 nN to 1 nN tightens the bound by a factor of 20. Increasing power from 50 W to 100 W tightens the bound by an additional factor of 2.

7 Implications

The present analysis converts EVRT from a purely predictive framework into a quantitatively constrained hypothesis. Two outcomes are possible:

- If future experiments detect signals within the allowed parameter space and with the required control-dependent scaling behavior, the model gains empirical relevance.
- If null results persist below the predicted range, the allowed parameter space is progressively eliminated.

This form of constraint analysis directly addresses the concern that EVRT is merely a scaling ansatz. While it does not provide a microscopic derivation, it makes the model quantitatively testable and places upper limits on the effective response parameter.

8 Limitations

This work is subject to several limitations:

- the underlying scaling relation remains phenomenological;
- field-energy scaling is represented by a reduced cavity-energy estimate;
- system-specific geometry, losses, and mode structure are not fully modeled;
- the bounds are illustrative and should be recalculated for any specific apparatus;
- independent experimental validation remains necessary.

These limitations do not invalidate the constraint approach. Instead, they define the next layer of refinement required for higher-fidelity comparison with experiment.

9 Conclusion

This work establishes upper bounds on effective electromagnetic stress-asymmetry behavior in resonant cavity systems using experimentally relevant sensitivity limits. For representative tabletop parameters, the effective response parameter is constrained at approximately $\chi_{\text{eff}} \lesssim 6 \times 10^{-17}$, with stronger bounds possible through improved force sensitivity, higher controlled power, and increased geometric asymmetry.

By framing EVRT as a constrained, testable hypothesis rather than an unconstrained scaling model, the analysis strengthens its scientific positioning and clarifies the requirements for meaningful experimental tests. Future work should focus on full-wave electromagnetic modeling, apparatus-specific constraint calculations, improved measurement sensitivity, and independent experimental validation.

In this sense, the present work represents the constraint stage of the EVRT research program, bridging predictive modeling and future experimental validation.

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